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CONFIGURATION OF THE BURIED SURFACE OF THE PRECAMBRIAN AND THE STRATIGRAPHY OF THE PALEOZOIC IN THE CENTRAL NEGEV: INDICATIONS FROM MAGNETO-TELLURIC MEASUREMENTS

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ABSTRACT

Magneto-telluric measurements along a 45 km N-S profile crossing the Nahal Paran basin, southern Israel, show a trough-like local deepening of the basement (Igneous Precambrian and/or Zenifim Formation) from 1.4-2.3 km to over 4 km. The sedimentary column is divided into four magneto-telluric units, broadly correlative to the Recent-Upper Eocene, the Hashefela Group (Middle Eocene-Senonian), the Judea Group (Turonian-Albian), and the combined Mesozoic-Paleozoic sandy formations. The latter may include Lower Paleozoic sediments such as are known from neighboring Jordan and Saudi Arabia (but not in Israel) or preserved sections of the 'Arad and Ramon groups, elsewhere denuded. A local fill is also possible. The trough is an ancient feature, apparently formed during a Paleozoic (pre-Negev Group), a Mesozoic (pre-Kurnub Group) and Neogene-Pleistocene phases. Limitations of the magneto-telluric methods are briefly discussed.

INTRODUCTION

A series of magneto-telluric measurements were conducted during 1970 in the central and southern Negev. The measured sites were distributed along an approximate north-south line, from the Meishar area to Har Zenifim across the Paran Valley (Figure 1). Altogether the line was some 45 km long.

The survey was an attempt to obtain geological data on the depth and shape of the buried surface of the Precambrian basement, and to locate prominent facies changes and truncations in the Paleo-Mesozoic sedimentary sequence.

The region was chosen because two deep boreholes located at the southern (Sinaf 1) and northern (Hameishar 1) edges of the surveyed area (Figure 1) penetrated a Paleo-Mesozoic sequence and reached Precambrian rocks (arkose sequence) at a relatively shallow depth (974 m and 1247 m). Thus the geological information could be used to help interpret the magneto-telluric data.

Earlier geophysical studies in the area include an aero-magnetic survey (Domzalski, 1967) and a gravimetric and magnetometric profile across the Paran Valley (Folkman, 1971), both conducted by the Institute for Petroleum Research and Geophysics.

The present survey was carried out in two stages. Seven sites were measured during the first stage (March 17-29). Since not all these measurements could be used in the interpretation, and in order to obtain clarification of the data, twelve more measurements were conducted (July 29 – August 7 and December 22-24).

The results showed basement* depth and revealed the presence of a thick sedimentary basin in the center of the area. The unusual thickness of the sedimentary column, which is nearly three times thicker here than elsewhere in the region raises stratigraphic problems that are the subject of this paper.

* With the present method, there was no possibility to differentiate between Precambrian sediments (Zenifim Formation) and the crystalline basement, and thus both are referred to as "basement".

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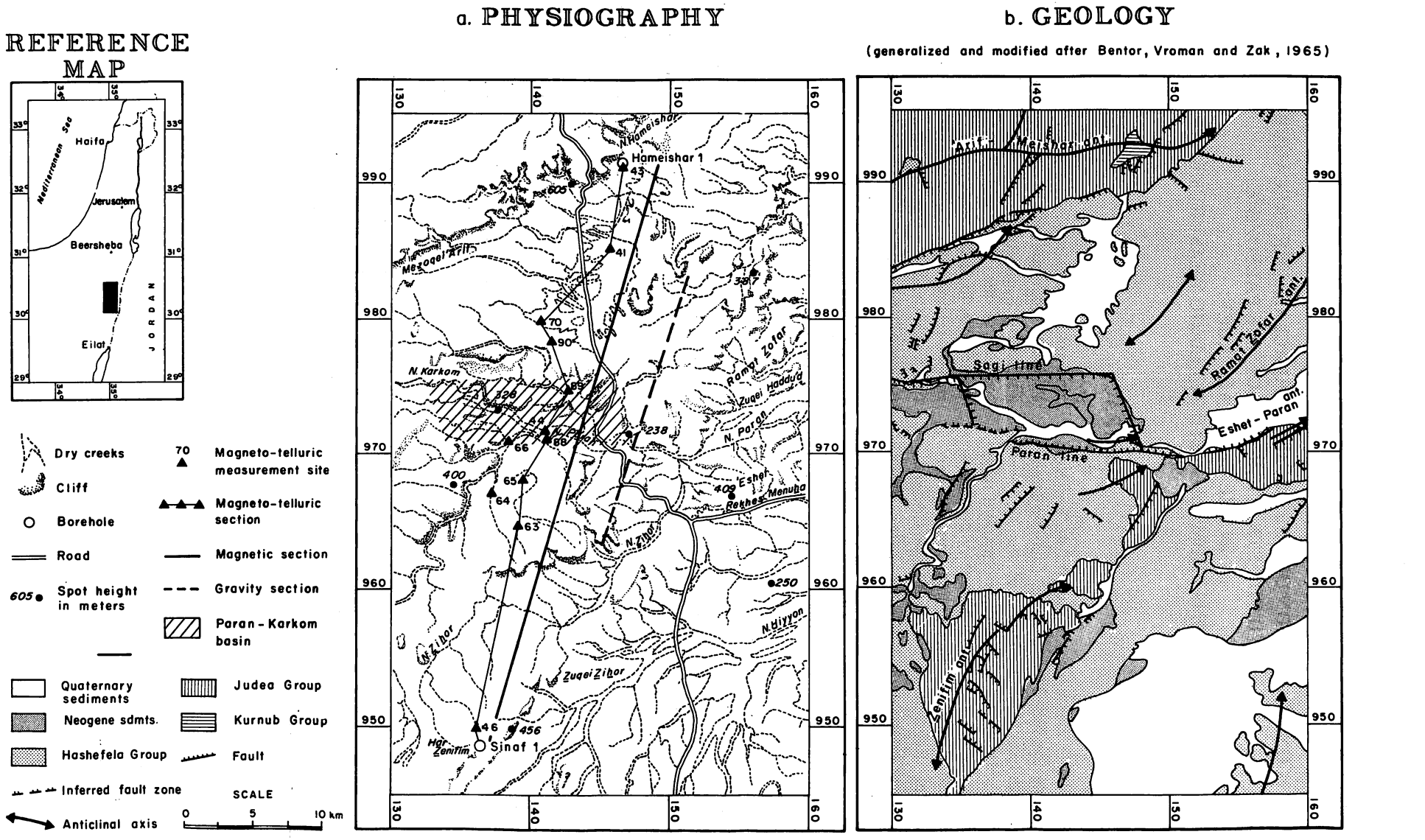


Figure 1: Location map of the Zenifim–Meishar area

THE MAGNETO-TELLURIC METHOD

The magneto-telluric method is an electrical geophysical technique for measuring resistivity as a function of depth. Its usefulness in exploration depends on differences in electrical properties exhibited by different rocks. In their simplest form the results of magneto-telluric measurements can be compared to a smooth electric log, giving resistivity *versus* depth information. Essentially, the magneto-telluric method involves the comparison of the simultaneously measured horizontal components of the magnetic and electric fields associated with the flow of telluric currents.

For a model assuming a horizontally layered earth, a quantitative interpretation is possible. Cagniard (1953) demonstrated that the ratio of two orthogonal tangential components, one of the electric field amplitude (Ex) and the other of the magnetic field amplitude (Hy) at a given frequency measured at the surface, depends on the resistivities down to the depth of penetration for that particular frequency. By determining this ratio at a series of frequencies, an apparent resistivity *versus* frequency curve may be calculated. In order to obtain an earth model of resistivities and thicknesses for the measured site, a method of quantitative interpretation based on curve matching is used whereby changes in the apparent resistivity curve are compared to theoretical curves of different resistivity contrast. By using a multi-layer case computer program, the theoretical curve for the computed model can be constructed.

In this study, only two orthogonal components, one of the telluric field amplitude (Ex) and one of the magnetic field amplitude (Hy) were measured. When measuring these two components of the magneto-telluric field, a horizontal isotropic layered model must be used. The accuracy of the results may be adversely affected if the actual geology at the measured site is of an anisotropic nature (presence of a fault, dike, or other type of discontinuity).

GEOLOGICAL BACKGROUND

Previous work

Geological data from the surveyed area, especially the deeper parts, are poor. Most were obtained during oil exploration, devoted mainly to structures in the central part and edges of the area, and from a study of iron mineralization in the Paran Valley. The area was mapped at 1:100,000 (Bentor and Vroman, 1951) and some stratigraphical studies were made, dealing chiefly with the exposed post-Lower Cretaceous sedimentary column.

The geological data cited below are chiefly based on the studies of Browne, Gwinn and Nasr (1941), Bentor and Vroman (1951), Sommer and Biro (1956), Sommer and Cohen (1956), Garfunkel and Horowitz (1966), Sakal (1967), Weissbrod (1969a) and Shraga (1971).

Previous geophysical data

Geophysical studies by Folkman (1971) include an interpretation of a gravity profile and a magnetic profile, approximately parallel to the present magneto-telluric section (Figure 2). The profiles were taken from an existing gravity map (unpublished Relative Bouger Anomaly Map of Israel) and an aeromagnetic map of Israel (Domzalski, 1967). Folkman concluded that the surface of the Judea Group in the Paran basin descends to a depth between 540 and 790 m below m.s.l. (depending on the density contrast postulated between the Judea Group and the Hashefela Group). The magnetic basement varies in depth from 1 km below m.s.l. in the Zenifim area to 3.7 km below m.s.l. in the Paran area. According to Domzalski's data and interpretation, the magnetic basement in this area is at a greater depth than anywhere in the south, and just south of Nahal Paran there appears to be an uplifted area.

Structure (Figure 1)

The area is one of sharp desert relief but with small differences (tens rather than hundreds of meters) in elevation. The landscape consists of flat reggan ("hammada") crossed by a few prominent ridges and dissected by the Riftward

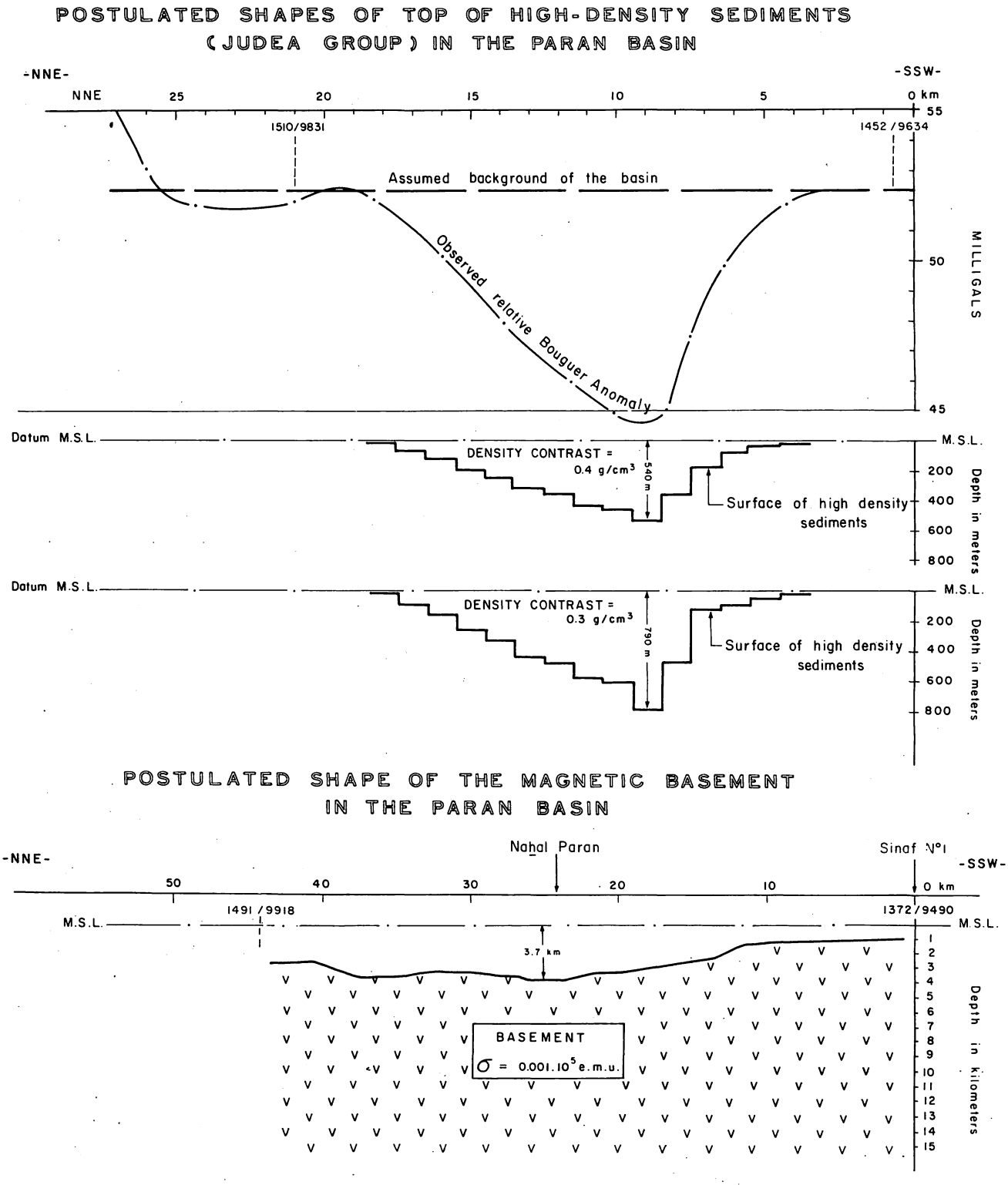


Figure 2: Gravity and magnetic profiles across the Paran basin (from Folkman, 1971)

wadi system. Structure is generally well expressed by the morphology: ridges are anticlinal, valleys are filled synclines or downfaulted depressions. Much of the secondary morphology is due to differential resistance to weathering, the marls, sandstones and unconsolidated gravels tending to form badlands, the hard carbonates and chert beds forming ledges and mesas.

Two main fold types are recognized in the area: the first are major anticlinal structures (such as Sinaf and 'Arif – Meishar) – asymmetric folds dipping steeply (45°-60°) towards the southeast or east and gently (5°) towards the west or northwest. The second type includes minor, gentle symmetrical folds (such as Ramat Ẓofar). An exception to these is the 'Eshet-Paran fold which is a faulted anticline with both flanks steep. The fold axes trend generally N 60°-80° E, but the Sinaf anticline trends approximately N-S and the 'Eshet-Paran chain, E-W.

The faults in the area trend mainly in two directions. The most prominent faults usually trend N 30°-75° E and can be followed for several km along the strike. These faults have vertical displacements of up to 600 m. The other trend is transverse to the fold axes and varies between N 15° E to N 55° W. This group of faults includes relatively short fractures with small vertical displacement. The main faults of the area are those of the Paran and Sagi lines, which continue into Sinai and are locally accompanied by ferruginous mineralization. The Paran fault, which was crossed by the magneto-telluric section is regarded by Sakal (1967) as a high-angle reverse fault with a maximal throw of 655 m.

Lithostratigraphy (from the surface downward)

Recent to Upper Eocene

Except for the Qeẓi'ot Formation this is a sequence of fluvial sediments that were deposited in basins, valleys and flood plains and which are mainly exposed in the present-day Paran-Karkom basin. The lower part of the sequence (Ḥaẓeva Formation) consists of conglomerates, detrital limestone, sandstone, marl and clays, whereas conglomerates comprise most of its upper part ('Arava Conglomerate and Hameishar Formation). The maximal known thickness of the sequence is 140 m but it is seldom fully present, due to unconformities and erosional gaps between the various units.

The Upper Eocene is represented by 22 m of the Qeẓi'ot Formation which belongs formally to the marine Saqiye Group. It occurs in a small exposure in the center of the area, composed of marl and shales with a few sandstone and limestone interlayers. According to Sakal (1967), the presence of this remnant suggests a bay in the Late Eocene sea which was regressing westward through the already existing Paran Valley. It is possible that the Qeẓi'ot Formation occurs elsewhere in the Paran Valley, covered by younger deposits. This may be supported by the interpretation of a gravity profile within the Paran Valley (Folkman, 1971) which indicates greater thickness (820-1,070 m) of the low-density sediments (Recent to Senonian) than what is known (~ 700 m) from the nearby exposure (Sakal, 1967). Note that Folkman's figure does not include the extra 280 m above m.s.l.

Hashefela Group (Middle Eocene – Senonian)

Sediments of this group are widely exposed in the surveyed area, mainly in the synclinal zones. Eight formations are recognized, consisting predominantly of chalky-marly, cherty and phosphoritic sequences. The thickness ranges from 450 to 650 m.

a. Middle Eocene sediments: These include four formations reaching a thickness of about 165 m. They consist mainly of white chalk, sometimes interbedded with thin layers of brown chert (Mor and Ḥorsha formations), and cliff-forming limestone (Niẓẓana and Matred formations).

b. Paleocene-Senonian sediments: These include four formations with an overall thickness of 300 to 460 m. The two upper (Ghareb and Taqiye formations) have a thickness of about 120 m and are composed mainly of marly-chalky sediments. The two lower formations (Mishash and Menuḥa) are here thicker than elsewhere in Israel. They consist mainly of white to

yellow chalk, dark chert and concretional limestone. The upper part of the Mishash Formation contains phosphorite beds.

Judea Group (Turonian-Albian)

Four formations of this group are exposed, mainly in the anticlinal ridges. The lower is the Ḥazera Formation consisting of limestone and dolomite alternating with marl. It is overlain by the Ora Shales consisting of clayey marl with some limestone. Overlying the Ora Shales are the Gerofit and Ẓiḥor formations consisting of limestone, chalk and some marl. The Judea Group forms the most resistant part of the sedimentary column in the area. Its thickness ranges from 400 to 600 m with the Ḥazera Formation thickening and the Ẓiḥor Formation thinning towards the north. The main facial change is the change from shales (Ora Shales) in the south to carbonates in the north (Derorim Formation). The Judea Group overlies the Ḥatira Formation unconformably (according to Garfunkel, 1970, the lower parts of the Ḥazera Formation interfinger with the Ḥatira Formation south of Paran Valley) and it is generally conformably overlain by the Menuḥa Formation.

Kurnub Group (Lower Cretaceous)

In the surveyed area, this group is represented by the Ḥatira Formation, which consists of variegated sandstones with siltstone and shale interlayers. The thickness ranges from 150 to 230 m. This unit was penetrated both in the Sinaf 1 and Hameishar 1 boreholes. Alternating variegated sandstone, clay and marl exposed in the 'Eshet anticline (east part of the area) were attributed by Sakal (1967) to the Lower Cretaceous. Garfunkel (1970), however considers this sequence to be part of the Ḥazera Formation (Judea Group).

'Arad Group (Jurassic)

Rocks of this group are not known in the surveyed area. According to Weissbrod (1969a) these sediments are completely truncated some distance south of Makhtesh Ramon. Lower Cretaceous sandstones unconformably overlie Triassic (Hameishar 1) to Cambrian (Sinaf 1) strata (Weissbrod, 1969a). However if the area was downwarped before the Jurassic had been eroded the latter should still be present in the subsurface of the Paran area. If this is the case then by extrapolating the original north-south rate of thinning we may expect a thickness of 200-250 m of dominantly clastic sediments (M. Goldberg, personal communication).

Ramon Group (Triassic)

Formations of this group occur in Hameishar 1. The upper parts (Moḥila and Saharonim formations) are missing and the remainder of the Triassic sequence is represented by 230 m of sandstones with thin shale interbeds (Gevanim Formation) overlying 65 m of hard carbonates (Ra'af Formation) and truncated by the Ḥatira Formation. Toward the south truncation has been deeper, resulting in complete removal of the Triassic sediments (Sinaf 1). However, again if the Paran area was down-faulted before the Early Cretaceous erosion, then part of the Triassic sequence (400-450 m according to Y. Druckman, personal communication) may be expected south of Hameishar 1.

Negev Group (Carboniferous – Lower Triassic)

Formations of this group occur only in the subsurface and were penetrated by Hameishar 1 borehole, at the northern end of the magnetotelluric section. The upper part of the group is composed mainly of carbonates, shales and sandstone while the lower part is almost pure sandstone. Sills and dikes intrude the lower part. The Negev Group was first defined, in the northern Negev boreholes (Weissbrod, 1969a) where four formations were recognized (Sa'ad, 'Arqov, Yamin, Ẓafir), on the basis of overall lithology and electric log patterns. Toward the south the facies of this group becomes sandy and the formations lose their identity. In Hameishar 1 the Negev Group overlies unconformably the Yam Suf Group while the contact with the Triassic sequence is conformable. The Negev Group in this borehole was previously (Weissbrod, 1969a) considered to overlie the Precambrian arkose sequence (Ẓenifim Formation). However, petrographic examination has shown that the feldspars in both the upper parts of the Ẓenifim

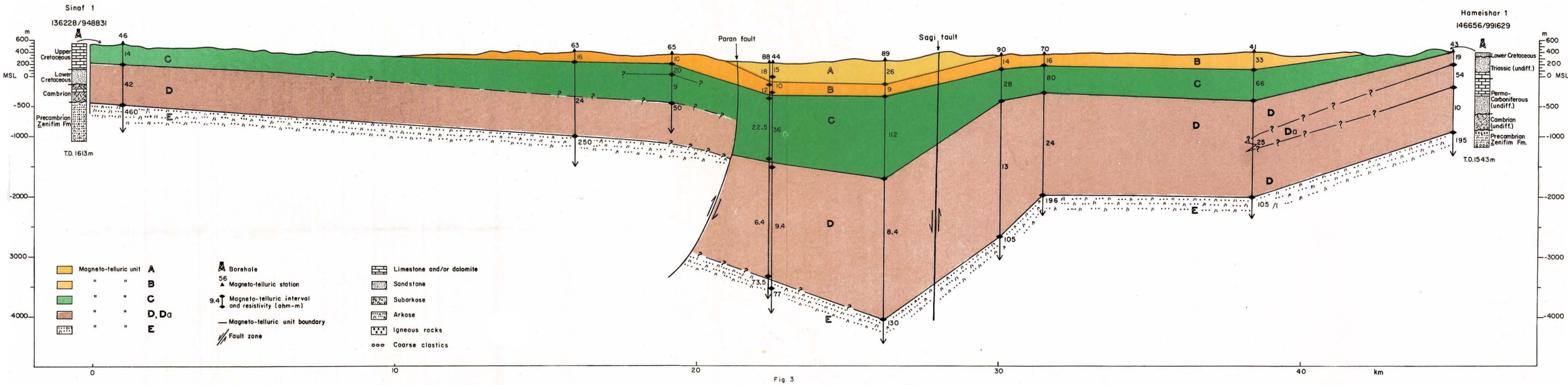


Figure 3: Magneto-telluric profile showing resistivity boundaries and proposed units

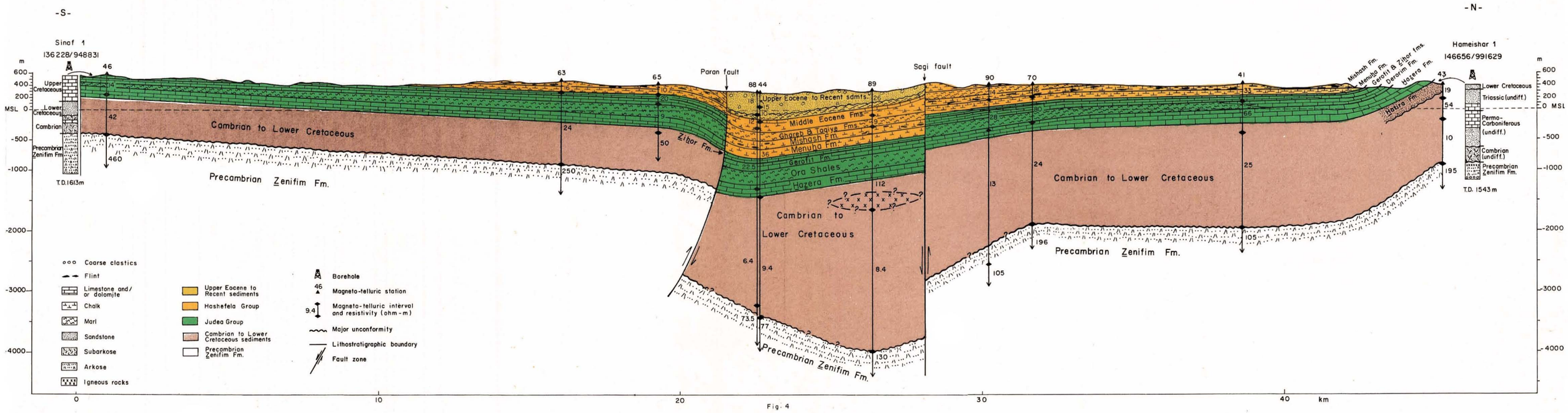


Figure 4: Geological cross-section showing upper lithostratigraphic units (based on regional outcrops and borehole data) and assumed configuration of magneto-telluric basement

arkose and the lower part of the Negev Group sandstone are dominated by microcline, a petrographic characteristic typical of the Cambrian sandstones in this area (Weissbrod 1969a, p. 4).

Accordingly, and due to the presence of some dolomite streaks in the same interval, the stratigraphy of Hameishar 1 is presently revised to 600 or 650 m of the Negev Group, 207 m of the Yam Suf Group and 297 m of Ẓenifim Formation, instead of the 714 m for the Negev Group and 410 m for the Ẓenifim Formation originally published. The Negev Group does not occur in Sinaf 1. Here, as in the Timna' – Elat exposures, the Negev Group was eroded and Lower Cretaceous sandstones directly overlie the Cambrian. However, the Negev Group probably continues a considerable distance south of Hameishar 1 and may still be expected in the Paran area.

Yam Suf Group (Cambrian)

Sediments of this group, otherwise known from outcrops in the Timna' area, to the south, were penetrated in both boreholes. In Sinaf 1 it is composed of 250 m of arkose overlain by 80 m of dolomite, shale and sandstone (Weissbrod, 1969a). In Hameishar 1 the group contains arkosic sandstone and micaceous shales with thin dolomite interlayers, some 210 m thick. The Yam Suf Group overlies unconformably the Precambrian arkose sequence (Ẓenifim Formation) and is in turn truncated by overlying Lower Cretaceous (Kurnub Group) sandstones in Sinaf 1, and by Permo-Carboniferous (Negev Group) sandstones in Hameishar 1.

Ẓenifim Formation (Precambrian)

This sequence was encountered in several of the deep Negev boreholes. It consists mainly of arkose, arkosic siltstone, conglomerates and some volcanics. The base of this sequence was not reached but it is assumed to overlie the crystalline basement (Weissbrod, 1969a). In Hameishar 1, 297 m of this sequence were penetrated, consisting of an upper basalt flow, arkose in the middle part and conglomerate at the base. In Sinaf 1, only arkose (600 m) was encountered. However, this arkose is not monotonous and shows petrographic differences between its upper and lower parts.

RESULTS OF THE MAGNETO-TELLURIC MEASUREMENTS

A total of 18 magneto-telluric measurements were made and 12 of these were amenable to interpretation (Table 1 and Figure 1). The rest of these measurements either failed due to technical problems or could not be clearly interpreted.

Table 1 – Results of magneto-telluric measurements

Site	Coordinates	Elevation (m.s.l.)	Interval (m)	Thickness (m)	Resistivity (ohm-m)	Magneto-telluric Unit
41	14575/98505	+365	0 - 270	270	33	B
			270 - 800	530	66	C
			800 - 2400	1600	25	D
			2400		105	E
43	14665/99163	+376	0 - 190	190	19	D
			190 - 550	360	54	Da
			550 - 1300	750	10	D
			1300		195	E
44	14085/97155	+283	0 - 230	230	15	A
			230 - 450	220	10	B
			450 - 1700	1250	36	C
			1700 - 3700	2000	9.4	D
			3700		77	E
46	13610/94990	+565	0 - 330	330	14	C
			330 - 1000	670	42	D
			1000		460	E
63	13925/96485	+450	0 - 175	175	16	B
			175 - 1400	1225	24	C+D
			1400		250	E
64	13715/96710	+325	0 - 190	190	13.5	B
			190 - 380	190	40.5	C
			380 - 1000	620	4	D
			1000		80	E
65	13035/96810	+415	0 - 140	140	10	B
			140 - 320	180	20	C
			320 - 800	480	9	C
			800		50	D
66	13855/97095	+300	0 - 230	230	15	B
			230 - 920	690	30	C
			920 - 1840	920	16	D
			1840		126	E
70	14085/97980	+395	0 - 175	175	16	B
			175 - 620	445	80	C
			620 - 2300	1680	24	D
			2300		196	E
88	14000/97135	+282	0 - 370	370	18	A
			370 - 600	230	12	B
			600 - 1600	1000	22.5	C
			1600 - 3520	1920	6.4	D
			3520		73.5	E
89	14270/97480	+305	0 - 390	390	26	A
			390 - 590	200	9	B
			590 - 1950	1360	112	C
			1950 - 4300	2350	8.4	D
			4300		130	E
90	14125/97850	+400	0 - 240	240	14	B
			240 - 750	510	28	C
			750 - 3000	2250	13	D
			3000		105	E

The interpreted magneto-telluric measurements indicate a series of five intervals with resistivities ranging from 4 to 460 ohm-m. The highest values generally occur in the deepest parts of the profiles.

Figure 3 presents a N-S cross-section (approximately 45 km) based on ten magneto-telluric profiles. By using the relative magnitudes of the resistivity values (and not the absolute values), five continual intervals or units (A, B, C, D, E) may be established appearing in this vertical order across the area. Each such unit is characterized by its specific range of resistivities and differs in this respect from overlying and underlying units.

A uniformed layered structure, rather than a structure consisting of wedges is assumed for the subsurface sections. The Cenomanian and younger units cropping out on the surface show no marked lithological lateral variations over the area. With regard to the lower part of the section (Cambrian to Lower Cretaceous), all these formations are known to be clastic irrespective of their age or stratigraphic position, and lithological discontinuities are not expected. A possible exception is the high-resistivity interval in unit D at site 43. For paleogeographic reasons, the possible presence of volcanics in unit D within the Paran basin (Figure 4) is allowed for.

The magneto-telluric units are believed to be generally correlative to the local lithologies, even though their boundaries do not coincide accurately with lithostratigraphic interfaces.

Lateral resistivity changes in each unit may be explained by the various factors which affect the resistivity in rocks — including porosity, fluid content and salinity, mineralization, formation temperature, degree of compaction and petrologic variability. An example for lateral changes in resistivity possibly due to differences in fluid regime is indicated in unit D within the deeper part of the basin. Generally resistivities of this unit range from 13 to 42 ohm-m nevertheless in the graben area it ranges from 6.4 to 9.4 ohm-m. This may be due to the presence of brines in an unflushed porous clastic sequence, preserved as a result of tectonic events.

Nevertheless, in evaluating such results three kinds of errors must be taken into account:

1. Errors in observation, which depend on the accuracy to which physical measurements can be made. In the present case these consist of the accuracy with which the electrode separations are measured (about ±1%), the accuracy of alignment of the electrode lines and of the magnetometers (about ±½ degree), and the accuracy with which the voltages are obtained by the recording system. Due to the high quality of instruments used in the field a general observation error of only ±2-3%, can be considered.

2. Errors inherent in the resistivity/thickness model, due to the uncertainty involved in matching a theoretical curve with a curve computed from field data (apparent resistivity curve) representing a horizontally layered situation. A statistical analysis of the errors involved in this phase was conducted and the error was found to be of a general order of ±10-12%. The error is much smaller (±2%) for the high-resistivity layer (Precambrian).

3. Errors of interpretation, caused by the postulation of a layered geometry even for cases of lateral change in resistivity. Since only two components of the magneto-telluric field were measured, lateral changes in resistivity (fault, dike, etc.) could not be interpreted otherwise.

It is not possible to calculate exactly the errors caused by these resistivity changes. However, theoretical studies by Keller and Frischknecht (1970) have shown how such changes affect the resistivities computed from a layered model.

The presence of a fault which causes lateral changes in resistivity, has no effect on the values computed for the high-resistivity side, but they may be much smaller than the true resistivity on the low-resistivity side, near the fault. At a lateral distance of about half the depth of substratum (high-resistivity) from the fault, the influence becomes very slight and is practically zero at a distance equal to the depth of the substratum.

For a dike or vertical mineralization zone the resistivity values are affected in the region close to the dike. The disturbance is smaller for a conducting dike than for a resistive one (Rankin, 1962).

In our magneto-telluric section (Fig. 3) the presence of the Paran fault may well be the cause of the lower basement resistivity values found near it (sites 88, 44) as compared to the basement elsewhere.

In summary, the depths and resistivities that were established (Table 1) must be regarded as approximate with allowance for a 20-25% composite error.

Due to the absence of subsurface geological data in the center of the survey area, the magneto-telluric profiles had to be correlated with the known stratigraphy north and south of the area. Two sites (43 and 46) were therefore located in the immediate vicinity of the Hameishar 1 (north) and Sinaf 1 (south) boreholes, both of which reached the Precambrian.

The correlation indicates that the high-resistivity unit E (deeper part of the profile) is correlative to the arkose of the Precambrian Zeni-

fim Formation (Weissbrod, 1969a). The top of this formation in the two boreholes coincides with the top of unit E in the nearby magneto-telluric profiles. The resistivity values of this unit (73.5-460 ohm-m), conform to values reported for crushed granite and consolidated arkose (100-500 ohm-m, Clark, 1966).

The other magneto-telluric units at sites 43 and 46 may be correlated with gross lithologies or even lithostratigraphic units from the nearby boreholes. Thus at site 46, unit C is correlated with the Judea Group carbonates penetrated in Sinaf 1, and unit D is correlated with the calstics of Lower Cretaceous and to the subarkoses of the Cambrian which they overlie. At site 43, unit D may be correlated with clastic sequence of the Lower Cretaceous, Triassic, Permo-Carboniferous and Cambrian, as it is encountered in Hameishar 1.

The intermediate zone of relatively high resistivities in unit D at site 43 (unit Da) may represent the section intercalated by Permo-Triassic carbonates, which are encountered in nearby Hameishar 1.

The overall correlations between magneto-telluric units and lithostratigraphic units are as follows:

Unit A is assumed to represent Neogene and younger sediments.

Unit B is assumed to represent the Hashefela Group.

Unit C is assumed to represent the Judea Group.

Unit D is assumed to represent the clastic rocks of the Cambrian to Lower Cretaceous sequence which despite its several stratigraphic hiatuses, consists dominantly of sandstones.

Unit E is assumed to represent the Precambrian Zenifim Formation (arkose and volcanics) and possible subcrops of the crystalline basement.

The thicknesses and boundaries of the magneto-telluric, and the interpolated lithostratigraphic units do not necessarily coincide. This must be attributed to any of the several kinds of systematic errors described above, i.e. inaccuracies of measurement, errors inherent in matching data curves with computed curves, and possible inadequacy of the isotropic layered model.

In most profiles the resistivities and the magneto-telluric units agree with the expected lithology. However, the following specific difficulties were noted:

1. At site 65, the top of the high-resistivity zone (unit E) could not be determined due to the uncertain resistivity values measured during low periods (ranging from 5 to 400 seconds).

2. At site 63 no clear differentiation between the magneto-telluric units C and D could be made and the correlation with the nearby sites is uncertain.

3. At site 89 an anomalous resistivity of 112 ohm-m was found in unit C. The reason may be, in addition to the possibilities of error, a possible local volcanic intercalation that affects the resistivity of the whole unit. At the same anomalous site, unit B is considerably thinner than what is known from other geological evidence (Sakal, 1967; Folkman, 1971). The reasons for the difference are not clear.

4. The resistivities of unit E generally exceed 100 ohm-m. An exception to this are sites 88 and 44, where resistivities of 73.5 and 77 ohm-m were recorded. Possible explanations for the low resistivities, besides error, might be proximity to the Paran fault, as explained above, and the geothermal anomaly and mineralization associated with the fault zone as reported by Eckstein (1973).

The information obtained by the magneto-telluric measurements, as interpreted (Fig. 3), brings to light two important features of the subsurface geology:

a. Depth and configuration of the surface of the Precambrian and the thickness of the overlying sedimentary column. The sedimentary column overlying the Precambrian in this trough is up to 4,200 m thick. Such a great thickness of sediments is not known anywhere else in southern Israel.

b. General distribution of resistivity units which conform clearly to gross lithostratigraphic units.

If the top of unit E represents, in fact, the surface of the Precambrian in all the magneto-telluric profiles, then a large basin or trough in the center of the surveyed area is indicated. This trough trends approximately E-W and is located beneath the present-day Paran Valley. A similar basement configuration with the same depth (Fig. 2) was independently found by the interpretation of a magnetic profile by Folkman (1971).

INTERPRETATION AND DISCUSSION

A geological cross-section along the magneto-telluric measurement sites is presented in Figure 4. The cross-section is hypothetic, based on interpolation between the boreholes and outcrops at the north and south ends of the section and outcrops along the way, on magneto-telluric results, and on Folkman's (1971) gravity and magnetic profiles.

The lower boundary of the cross-section is taken as the top of magneto-telluric unit E which is believed to represent the surface of the Precambrian (i.e., Ẓenifim arkose and igneous basement). The sedimentary column between the Judea Group and the Precambrian is not differentiated due to insufficient data. The various possibilities, however, are discussed below.

A fault on the northern side of Naḥal Paran (Bartov, 1973) is also indicated from the fact that the Precambrian unit E shows abrupt deepening in this area (between measurement sites 90 and 89). This fault is the continuation of the so-called Sagi line (Bentor and Vroman, 1954).

The Paran-Karkom basin appears thus to be bounded on the north and south by two parallel east-west faults (considered "strike-slip faults" by Bentor and Vroman, 1954 and by Bentor, 1966, and which according to Bartov (1973) have a dextral shear). Actually, the Paran-Karkom basin is a rhomb-shaped graben which is bounded by faults also on its eastern and western sides (see Fig. 1b). This kind of faulting, which was suggested at the time by R. Freund (personal communication) has been lately proven by Bartov (1973). In this graben Upper Eocene, Neogene and younger sediments have accumulated and were preserved.

The Paran and the Sagi faults are related either to a Late Eocene to Oligocene faulting phase (Bentor and Vroman, 1951), or to an Oligocene to Early Miocene phase (Picard, 1951) and a post Ḥaẓeva Formation stage (Bartov, 1973). Any of these proposed phases of faulting would account for the differences in thickness between Upper Eocene – Recent sediments within the Paran-Karkom basin and on its margins, but not for the enormous differences in thickness (over 1,100 m) between the pre-Judea Group sediments, represented by magneto-telluric unit D, within the basin and on its margins. It is therefore reasonable to assume that the faulting along the present Paran and Sagi lines, is not only a young feature but that those were active lines of dislocation since Mesozoic or even Paleozoic times. It should be noted that Sakal (1967) and Garfunkel (1970) maintain that the Paran line was active at least since Upper Cretaceous times.

Another category of evidence for ancient dislocation are the increasing distances of displacement of stratigraphic surfaces. Older surfaces are more displaced than younger ones, which were nonexistent at the time of the ancient dislocation. Thus the top of the basement (unit E) is approximately displaced 1,450 m at the Paran line and about 900 m at the Sagi line; the base of the Judea Group (unit C) is displaced 700 m and 400 m respectively, and the base of the Eocene formations is displaced 400 m and 300 m (Figs. 3 and 4).

Although the time of beginning of the tectonic activity is not known, a reasonable approximation can be made by reconstructing the stratigraphic sections on both sides of the faults. It is assumed that tectonic activity along these lines began during one of the major phases of uplift and faulting before the Tertiary-Neogene phase. In southern Israel, two such phases are known, and may be considered: a phase preceding the Carboniferous, and a phase preceding the Cretaceous. Each of these was followed by erosion on a regional scale, today marked by

great stratigraphic gaps. The possibility that the Paran-Karkom basin was initially formed during one of these phases, raises several consideration:

a. Post-Jurassic – pre-Cretaceous faulting and Neogene rejuvenation

If the trough was first formed at the time of the pre-Cretaceous regional uplift, we should expect differences in the thickness of the Lower Cretaceous sediments within and outside the trough, but not in later or earlier deposits. Actually one observes differences in the thickness of several other units on both sides of the faults, but these are due to the regional north-south trend of thinning against the Arabo-Nubian massif and to the general levels of regional erosion (Fig. 5). However, the differences of more than 1,400 m in thickness of the total section between the trough and areas north of it must, under the above assumption, be attributed to fill of the Lower Cretaceous Kurnub Group. Even if we assume part of this thickness to be due to an uneroded remnant of the Lower Mesozoic 'Arad and Ramon groups, the Lower Cretaceous Kurnub Group would still be twice or even more as thick within the trough than outside it.

The development of such a situation requires the formation of a very deep tectonic depression – over 1,200 m – over a very short time: after the end of the Jurassic, and before the regional truncation of the Early Cretaceous started. It requires also the availability of large amounts of clastics, as well as the means to transport them rapidly. These are anomalous conditions but well within the limits of geological possibility, as demonstrated by today's nearby Rift Valley. But, as the case is, the inferred composition of the sedimentary column north of the trough leads to the conclusion that at least part of the anomalous thickness must be attributed to Paleozoic sediments, probably unknown yet in Israel. This is concluded from a careful estimation of thickness in the sequence north of the Sagi line which indicates that even outside the trough the thickening cannot be attributed to the Kurnub Group only. Here sediments reach a total thickness of 2,300 m of which 700 m are post-Kurnub sediments known from outcrops. As for the deeper formations, there is no reason to believe that the Yam Suf Group is thicker here than in Sinaf 1 or in Hameishar 1, i.e., 250-330 m. The Negev and Ramon groups are known to wedge out southward, and are probably thinner here, also due to the regional truncations, than in Hameishar 1, i.e., 650 m. The 'Arad Group is already absent in Hameishar 1 and will therefore not appear south of it. Thus, even after adding 250 m of Kurnub Group sediments (their thickness at the Hameishar 1 and Sinaf 1 localities), some hundreds of meters, between 420 and 450 at least, have to be accounted for. In other words, the total thickness of the sedimentary column north of the Sagi line exceeds the maximum total known for Cambrian or post-Carboniferous formations. There must therefore be a considerable thickness of Paleozoic sediments underlying the Negev Group between Hameishar and Sinaf boreholes, reaching at least 450 m at sites 70-41 and probably accounting for much of the added thickness within the Paran-Karkom trough. This hypothetical situation is shown in Figure 5. These Paleozoic sediments could well be part of the Ordovician to Lower Carboniferous formations which occur in northwestern Saudi Arabia (Ordovician to Devonian), southern Jordan (Ordovician to Silurian) and southwestern Sinai (Lower Carboniferous),and which have undergone truncation elsewhere in southern Israel.

South of the trough on the other hand, the known thickness of the Yam Suf Group, even with the probable absence of the Ramon and most of the Negev groups, can account for the observed thickness without any anomalous thickening of the Kurnub Group.

The differences in the present thicknesses of this unknown Paleozoic between the trough and the area north of it could not have existed, if conditions in the area were uniform till the – supposed – formation of the Paran-Karkom trough in post-Jurassic times. These differences would have been obtained if depression of the trough had started earlier, shortly after their de-

position and before deposition of the Late Paleozoic Negev Group.

The abrupt thickening of the Paleozoic section between measurement sites 70, 90 and 89 suggests that, beside the ancient Sagi fault line which is seen even today in the field, the great depth of the trough may have been caused by a number of additional step faults not visible on today's surface.

b. Paleozoic faulting and Neogene rejuvenation

If tectonic activity along the Paran and Sagi lines had already begun in the pre-Upper Carboniferous uplift phase, then those parts of the sedimentary column within the trough that represent excess thickness as compared with the nearby two boreholes, must be attributed to the Paleozoic.

Taking into account the regional truncations of all pre-Cretaceous units, and their tendency to wedge out to the south, this excess thickness reaches at least 1,650 m and may be partly attributed to the Lower Paleozoic sequence (Ordovician to Lower Carboniferous) not yet known in Israel but known to have been deposited in Jordan (e.g. Massive Whitish Weathered Sandstone to Nautiloidea Sandstone sequence, Bender, 1968), Saudi Arabia (Saq, Tabuk and Jauf formations, Powers *et al.*, 1966) and southwestern Sinai (Um Bogma and Ataqa formations, Weissbrod, 1969b).

It is possible that due to its downwarping the Paran-Karkom basin remained, throughout the period preceding the regional deposition of the Negev Group, a local accumulation basin in which sediments were rapidly accumulated as the nearby areas were uplifted and eroded. This hypothetical situation is shown in Figure 6.

North and south of the trough the interpreted section is the same as that described for the pre-Cretaceous faulting possibility.

Possibility (b) appears the more probable since we have evidence for excessive Paleozoic thickness even outside the trough. It also offers a reasonable scheme for vertical displacement of the various stratigraphic surfaces.

However, it seems unlikely that a tectonic event like the pre-Cretaceous uplift should leave an existing trough feature unaffected. The events which caused filling of the Paran-Karkom basin are therefore probably the sum of total of Late Paleozoic, pre-Lower Cretaceous and Neogene pulses of displacement.

Accordingly, the geological section within the trough will be in general similar to that presented in Figure 5, with somewhat different thicknesses of the same stratigraphic units. The excess sedimentary thickness is thus mostly within the Paleozoic (unknown Lower Paleozoic, and possible overlying pre-Negev Group fill), partly in the Mesozoic Ramon, 'Arad, and Kurnub groups, and in the Neogene-Quaternary fill.

The pre-Judea Group sediments are considered to be mostly clastic for the following reasons:

1. The Lower Paleozoic sequence in northwestern Saudi Arabia and southern Jordan is mostly clastic, consisting mainly of sandstone with subordinate shale.
2. Borehole samples of sediments of the Negev Group show a north-south trend toward an increasing by sandy facies (Weissbrod, 1969a).
3. Resistivity values of this interval within the trough are very low (6.4 to 9.4 ohm-m) and suggest very porous sediments.

Carbonate rocks may occur within the Triassic or even Jurassic sections of this area, if present. These are probably thin, and their resistivities would be masked by those of the thick clastics.

It might be argued that basalt flows or other volcanics could account for part of the thickness in the trough, especially at the base of the Kurnub Group. Flows are known from this part of the section in the Makhtesh Ramon and Har 'Arif exposures, and in Hameishar 1 borehole. Resistivity values, however, indicate no such intercalations (with the possible exception at site 89). If flows or dikes are present they may well be too thin to be expressed in the magnetotelluric record.

The Paran line, as a line of tectonic discontinuity, extends westward into Sinai. Bartov (1973) observes it in the 'Arif e-Naqa, Gebel Kherim and Gebel Bruk faults. The line may have continued also eastward into Jordan.

Following the assumption that a major strike slip movement occurred along the 'Arava Valley, the continuation of the Paran line must be looked for about 105 km to the north (Freund *et al.*, 1970). The Gebel Shihan – Gebel Sawara fault on the eastern side of the Dead Sea may be part of this eastern continuation.

Troughs similar to the Paran-Karkom basin may also be expected along other localities on this line. A trough which was formed or renewed in Neogene or later times will be morphologically apparent (containing uneroded sections of the Ḥaẓeva Formation) while troughs that were formed by ancient tectonic movements may be discovered only by careful subsurface observations.

CONCLUSIONS

1. The present magneto-telluric survey in the Paran-Karkom basin, indicates the presence of a 2,500 m deep, trough formed by the Precambrian basement (igneous rock and Ẓenifim arkose). The trough is bordered by the east-west Paran and Sagi fault lines and coincide roughly with the Neogene-Quaternary depression in the area.

2. Sediments within the trough reach a thickness of 4,200 m. This is 1,800 m and 2,800 m thicker than on the respective southern and northern sides. The excess sediments must belong to formations that were either eroded or not deposited outside the trough.

3. Part of the excess thickness is known to be of Neogene-Quaternary age. The thickness values of the magneto-telluric units suggests, however, that the Judea Group in the trough area is not significantly thicker, so that the surplus sediment thickness was apparently deposited in pre-Judea Group times.

4. The trough was formed probably during one or several of the known tectonic phases in the region. Possibilities are the Paleozoic (pre-Negev Group) phase and the Late-to-post-Jurassic (pre-Kurnub Group) phase.

5. Taking into account regional truncations and trends of thinning, as well as stratigraphic relations south and north of the trough (Sinaf 1 and Hameishar 1 boreholes), it is most likely that the first movement occurred during the Paleozoic phase and a second pulse during the Late to post-Jurassic phase.

6. The Paleozoic sediments apparently include, beside the Yam Suf (Cambrian) and Negev (Upper Carboniferous-Triassic) groups known in the area, a thick intervening section of Ordovician to Lower Carboniferous formations, correlative to similar formations in the adjacent regions. A certain thickness of local trough-fill is also expected.

7. Downwarping during the Late to post-Jurassic tectonic phase would have caused the preservation of the 'Arad, Ramon and probably part of the Negev groups from the effect of regional denudation, adding their thickness to the sediment sequence in the trough. Basal beds of the Kurnub Group may add to the thickness of this unit in the trough.

8. The above conclusions are highly hypothetical and are based first and foremost on magneto-telluric information on basement depth and configuration. The introduction of new equipment, enabling the measurement of five field components and extension of the measuring periods, would greatly enhance accuracy of the results and the dependability of interpretation. Borehole and/or seismic data are needed in order to confirm the present or any other hypothesis.

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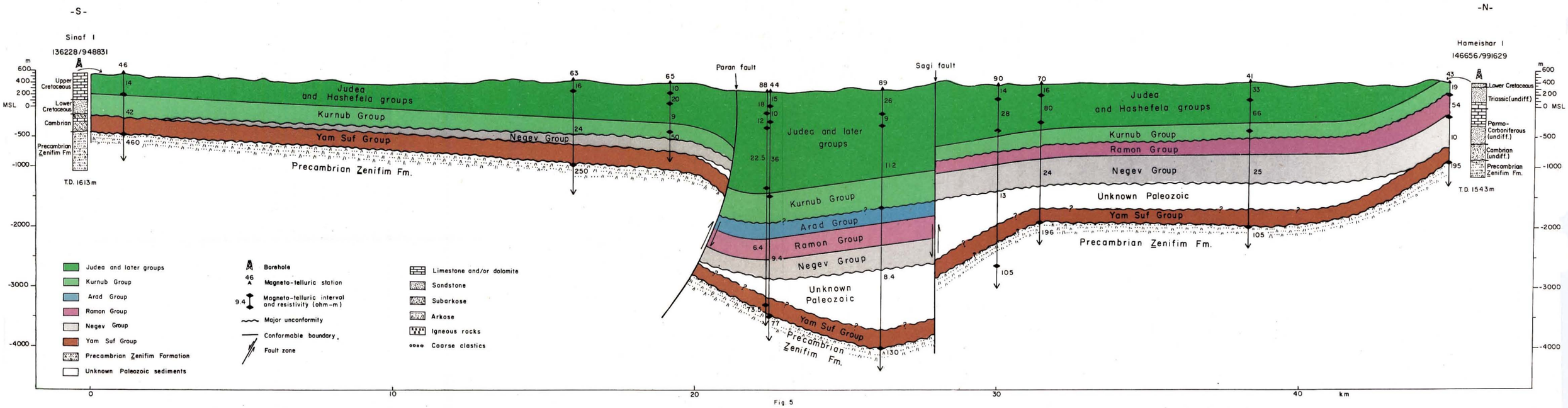


Figure 5: Reconstructed cross-section of the Paran-Karkom basin according to possibility (a): pre-Cretaceous faulting and Neogene rejuvenation

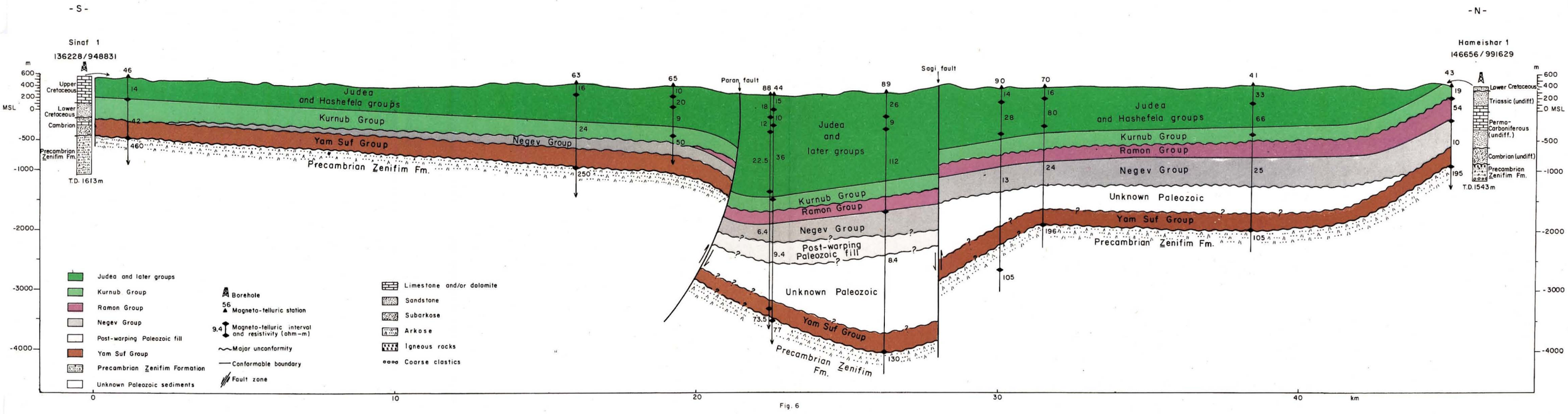


Figure 6: Reconstructed cross-section of the Paran-Karkom basin according to possibility (b): pre-Carboniferous faulting and Neogene rejuvenation

מדינת ישראל

משרד הפיתוח
המכון הגיאולוגי

צורת גג הפריקמבריום והסטרטיגרפיה של הפליאוזואיקון בתת-הקרקע של הנגב המרכזי: עדות מנתונים מגנטו-טלוריים

ט. ויסברוד
המחלקה למינרלוגיה, פטרוגרפיה ומחצבים
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א. קלנג
המכון למחקרי נפט וגיאופיסיקה
חולון

ירושלים, אייר תשל"ד

בולטין מס. 60